

PMM3 vs MLE-TN: Comparison of Semiparametric and Parametric Estimation of Linear Regression Parameters under Two-piece Normal Distributed Errors

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Abstract. This paper addresses the problem of estimating the parameter vector of a linear regression model under symmetric non-Gaussian errors following a Two-piece Normal (TN) distribution. Three methods are compared: Ordinary Least Squares (OLS), Maximum Likelihood Estimation for the TN distribution (MLE-TN), and the third-power Polynomial Maximization Method (PMM3). MLE-TN is fully parametric; PMM3 is semiparametric, using only the 2nd, 4th and 6th-order moments. Theoretical analysis shows that PMM3 achieves up to a 95.5% variance reduction compared to OLS at $\lambda = 5$ (the TN shape parameter), while MLE-TN provides up to a 28-fold advantage for large λ . A Monte Carlo study confirmed that the empirical efficiency coefficient $\mathcal{G}^3$ converges to the theoretical value for $n = 100$. Under misspecification PMM3 never falls below OLS (ARE from 1.02 to 2.68), whereas MLE-TN can be less efficient than OLS. On real-world data (*Iris versicolor*), PMM3 achieves a 43% bootstrap variance reduction and improves LOO-CV by 2.3% over OLS. An empirical efficiency threshold for PMM3 is established at $\gamma_4 \leq -0.7$.

Keywords: Polynomial Maximization Method, Linear Regression, Two-piece Normal distribution, Maximum Likelihood Estimation, Estimation Efficiency, Robustness, Monte Carlo.

1 Introduction

The Ordinary Least Squares (OLS) method [6] is the Best Linear Unbiased Estimator under Gaussian errors, but its efficiency depends on the shape of the error distribution [2]. When normality is violated, OLS loses its asymptotic optimality, whereas estimators exploiting higher-order moments may achieve lower variance.

In practical applications, including metrology, medicine, econometrics, and the natural sciences, measurement errors frequently follow symmetric but non-Gaussian distributions. Platykurtic distributions ($\gamma_4 < 0$) commonly arise in measurements with

a limited operating range or when several uniformly distributed error sources are aggregated.

One important family of such distributions is the Two-piece Normal (TN) distribution, introduced in [9] and further investigated in [17]. The TN distribution covers a broad spectrum of symmetric shapes, ranging from an approximately Gaussian form at ($\lambda \approx 0$) to a strongly bimodal form at ($\lambda \gg 1$), making it a natural benchmark for studying symmetric platykurtic distributions with $\gamma_4 \in (-2, 0)$.

Three approaches can be distinguished. The parametric approach (MLE-TN) jointly estimates all parameters $(\beta|\eta|\lambda)$, attains the Cramér–Rao lower bound under correct specification, but is sensitive to misspecification [7], [23]. The semiparametric approach (PMM3) uses only the 2nd, 4th and 6th-order moments and does not require the functional form of the error distribution. Nonparametric and robust approaches include Least Absolute Deviations (LAD) regression [16], quantile regression [11] and adaptive MLE [20].

The study asks when, and by how much, PMM3 outperforms MLE-TN and OLS.

1.1 Literature Review

In [2], a general expression for the asymptotic relative efficiency of OLS was derived in terms of the cumulants of the underlying distribution. Subsequently, [21] systematically investigated parameter estimation in linear regression models with symmetric non-Gaussian errors. The studies reported in [1] and [5] confirmed that, for $\gamma_4 < -0.5$, classical OLS may perform worse than the optimal estimator. Robust regression techniques [7] and LAD regression [16] are primarily designed for leptokurtic ($\gamma_4 > 0$) distributions. In contrast, for platykurtic ($\gamma_4 < 0$) distributions, none of the standard methods can be regarded as optimal. In [15], a semiparametric regression framework with an unspecified error distribution was developed and shown to outperform MLE when the assumption of normality is misspecified; Kim [10] combined semiparametric efficient scores with parametric or nonparametric error models chosen by the shape of the errors. In [17], both Bayesian and frequentist approaches to linear regression with TN-distributed errors were proposed, and MLE estimation was investigated using L-BFGS-B optimization. Furthermore, [13] developed an objective Bayesian methodology for the broader class of two-piece distributions. Both studies emphasize that, under model misspecification, parametric approaches may be inferior to semiparametric alternatives [23].

Recent work has extended two-piece models in several directions: an asymmetric double normal distribution for bimodal and skewed regression data [18], a modified maximum likelihood estimator for censored regression with two-piece generalized t errors [14], two-piece semiparametric quantile regression for right-censored data [3], two-piece scale mixtures of normals for robust mixture regression [26], and skew-normal errors with a nonparametric scale-mixing distribution that reduces the risk of misspecification [19]. All of these methods are likelihood-based and specify the error

density fully or in part; none offers a moment-based estimator for symmetric platykurtic errors.

As for the Polynomial Maximization Method (PMM), it was developed by Kunchenko and Lega [12]; the efficiency coefficient of PMM3 was derived in [22].

In [25], PMM was extended to linear regression with asymmetric errors (PMM2, $g_2 = 1 - \gamma_3^2 / (2 + \gamma_4)$) and in [24], the first application of the PMM3 to linear regression with symmetric non-Gaussian errors of general form was presented. In [8], a normality test based on sample skewness and kurtosis was proposed; on its basis residuals are routed to OLS (Gaussian), PMM2 (asymmetric) or PMM3 (symmetric platykurtic).

1.2 Scientific contribution of the study

A direct comparison of PMM3 and MLE-TN within the TN framework has not been carried out before, although both methods have been studied separately. The contributions of this work are as follows:

(I) analytical expressions for $g_3(\lambda)$, $ARE(MLE-TN)(\lambda)$ and the PMM3 vs MLE-TN efficiency curve; (II) Monte Carlo validation for $\lambda \in \{0.5, 1, 1.5, 2, 3, 5\}$, $n \in \{46, 100, 200, 500\}$ with $M = 500$ replications; (III) robustness: PMM3 stays above OLS under misspecification (ARE 1.02–2.68), whereas MLE-TN can fall below it; (IV) 100% Newton-Raphson convergence in all 24 Monte Carlo scenarios; (V) an empirical efficiency threshold $\gamma_4 \in -0.7$ based on seven real-world datasets; (VI) an application to Iris versicolor ($n = 50$, $\gamma_4 = -0.966$), with a 43% variance reduction.

2 Materials and methods

2.1 Linear regression model

Consider the following linear regression model:

$$y_v = x_v^T \beta + \varepsilon_v, \quad v = 1, \dots, n, \quad (1)$$

where y_i denotes the response variable, $x_i \in \mathbb{R}^p$ is the vector of regressors, $\beta \in \mathbb{R}^p$ is the vector of unknown regression parameters, and ε_i are independent and identically distributed random errors.

Assumption. For PMM3: the error term satisfies $\varepsilon_v \sim$ a symmetric distribution with $E(\varepsilon) = 0$ and possesses finite moments up to the sixth order. **For MLE-TN:** the error term follows the Two-piece Normal distribution, $\varepsilon_v \sim TN(0, \eta, \lambda)$.

2.2 Two-piece Normal (TN) distribution

The Two-piece Normal (TN) distribution with zero location parameter, scale parameter $\eta > 0$, and shape parameter $\lambda > 0$ has the probability density function [17]:

$$f(z | \lambda) = \frac{1}{\sqrt{2\pi}} \cdot \exp\left(-\frac{z^2}{2} - \frac{\lambda^2}{2}\right) \cdot \cosh(\lambda z), \quad z \in \mathbb{R} \quad (1)$$

The principal properties of the TN distribution are as follows: $\lambda \rightarrow 0$ – the distribution approaches the standard normal distribution $N(0, 1)$; $\lambda > 1$ – the distribution becomes bimodal, exhibiting two symmetric modes; γ_4 is strictly negative for all values of λ , indicating a platykurtic distribution.

The normalized excess kurtosis is given by:

$$\gamma_4 = \frac{\mu_4}{\mu_2^2} - 3 < 0 \quad \text{for all } \lambda > 0. \quad (1)$$

2.3 MLE-TN: Maximum likelihood estimation

The MLE-TN approach jointly estimates the parameter vector (β, η, λ) by maximizing the log-likelihood function:

$$\ell(\beta, \eta, \lambda | y) = \sum_{v=1}^n \log f\left(\frac{y_v - x_v^T \beta}{\eta} | \lambda\right) - n \log \eta \quad (1)$$

The optimization procedure is performed using the L-BFGS-B algorithm with the following initial values: $\beta_0 = \hat{\beta}_{OLS}$, $\eta_0 = \exp(\log \text{SD}(\hat{\epsilon}_{OLS}))$, $\lambda_0 = 1.0$. MLE-TN attains the Cramér–Rao bound under correct specification, but it estimates $p + 2$ parameters and may be numerically unstable for small sample sizes.

2.4 PMM3: Polynomial Maximization Method (S = 3)

The PMM3 estimator evaluates only the regression parameter vector β by employing a moment-corrected score function. The algorithm consists of the following steps: compute the initial estimate $\beta^{(0)} = \hat{\beta}_{OLS}$ and evaluate the sample central moments of the residuals m_2, m_4, m_6 , then the correction coefficient is estimated once according to

$$\kappa = \frac{m_6 - 3 m_4 m_2}{m_4 - 3 m_2^2} \quad (1)$$

If the corresponding condition is violated $|m_4 - 3 m_2^2| < 10^{-10}$, the algorithm returns the Gaussian fallback solution $\hat{\beta}_{OLS}$.

The parameter vector is updated according to the Newton-Raphson procedure: $Z_{1v} = \varepsilon_v (\kappa - \varepsilon_v^2)$; $Z = X^T Z_{1v}$; $J = X^T \text{diag}(3\varepsilon^2 - \kappa) X$; $\beta \leftarrow \beta - J^{-1} Z$. The iterations continue until the prescribed convergence criterion is satisfied: $\|Z\|_2 < 10^{-6}$. In practice, convergence is typically achieved within 3-5 iterations. To ensure numerical stability, if the specified divergence condition is met, $\|\beta - \hat{\beta}_{OLS}\| > 10$ the algorithm returns the initial estimate $\hat{\beta}_{OLS}$.

The theoretical efficiency coefficient of PMM3 is

$$g_3 = 1 - \frac{\gamma_4^2}{6 + 9\gamma_4 + \gamma_6}, \quad (1)$$

where $\gamma_4 = \mu_4 / \mu_2^2 - 3$, $\gamma_6 = \mu_6 / \mu_2^3 - 15(\mu_4 / \mu_2^2) + 30$ – normalized fourth and sixth-order cumulants, estimated directly from the sample.

2.5 Asymptotic normality of PMM3

Under the standard regularity conditions, the PMM3 estimator is consistent and

asymptotically normally distributed $\sqrt{n}(\hat{\beta}_{PMM3} - \beta) \rightarrow N\left(0, g_3 \sigma^2 (X^T X / n)^{-1}\right)$. Its asymptotic covariance matrix is reduced by a factor of $(1/g_3)$ relative to that of the OLS estimator: $Var(\hat{\beta}_{PMM3}) = g_3 \cdot Var(\hat{\beta}_{OLS})$.

For the Gaussian case ($\gamma_4 = 0$) $g_3 = 1$ PMM3 is asymptotically equivalent to OLS. For platykurtic distributions ($\gamma_4 < 0$) $g_3 < 1$ PMM3 is more efficient than OLS. The consistency of PMM3 depends only on the assumption of zero-mean errors and does not require specification of the exact distributional form. A more detailed derivation is provided in [12] and [25].

2.6 Theoretical characteristics of the TN distribution

The shapes of the TN density are shown in the left panel of Fig. 1 and the theoretical efficiency curve $g_3(\lambda)$ in the right panel. For the six shape values used in this study, $\gamma_4 = -0.080, -0.500, -0.959, -1.280, -1.620$ and -1.849 with $g_3 = 0.9988, 0.9286, 0.6574, 0.3868, 0.1490$ and 0.0454 for $\lambda = 0.5, 1.0, 1.5, 2.0, 3.0$ and 5.0 (variance reduction of 0.1% to 95.5%). The distribution is bimodal for $\lambda > 1$, that is for every $\lambda \geq 1.5$ in this set.

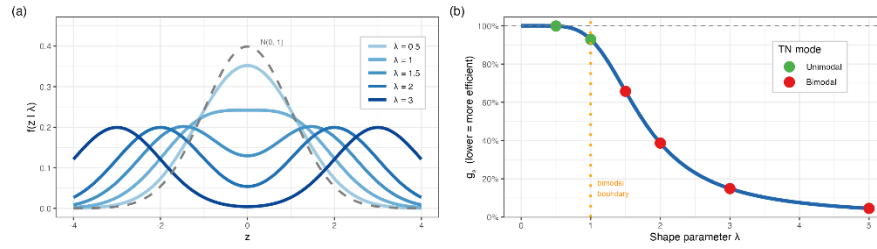


Fig. 1. Two-piece Normal model. Left: TN densities for $\lambda \in \{0.5, 1, 1.5, 2, 3\}$ (standard normal dashed). Right: theoretical efficiency g_3 versus λ ; the dotted line marks the bimodality boundary $\lambda = 1$.

3 Experiments

Monte Carlo design. The data-generating model is defined as:

$$y_v = 1 + \beta_1 x_v + \varepsilon_v, \quad \varepsilon_v \sim TN(0, 1, \lambda) \quad (1)$$

The simulation parameters: number of replications $M = 500$; sample sizes $n \in \{46, 100, 200, 500\}$; shape parameters $\lambda \in \{0.5, 1, 1.5, 2, 3, 5\}$; random seed = 20260312.

Metrics: Bias = $\hat{\beta}_1 - 1$; Var($\hat{\beta}_1$); MSE = Bias² + Var; ARE = Var(OLS)/Var(method); $g_3 - \text{emp} = \text{Var(PMM3)}/\text{Var(OLS)}$; relative deviation from theory $g_3 = (g_{3\text{emp}} - g_{3\text{theor}}) / g_{3\text{theor}} \times 100\%$.

Real-world data. A subset of the classical Fisher Iris dataset [4] corresponding to the species Iris versicolor was used for empirical evaluation. The dataset contains $n = 50$ observations.

$$\text{Sepal.Length}_v = \beta_0 + \beta_1 \cdot \text{Sepal.Width}_v + \varepsilon_v \quad (1)$$

OLS Residual Diagnostics: $n = 50$; $\hat{\gamma}_3 = 0.273$ (indicating only negligible asymmetry); $\hat{\gamma}_4 = -0.966$ (indicates pronounced platykurticity); $span / SD \approx 3.58$ – suggesting a restricted measurement range.

The corresponding theoretical efficiency is: $g_{3\text{theor}} = 0.483$ which corresponds to a 51.7% reduction in variance. Based on the residual diagnostics: $\gamma_4 = -0.966 < -0.7$ PMM3 is recommended.

Misspecification scenarios. Robustness was assessed on four non-TN error laws (uniform, triangular, logistic and Student's t with 10 degrees of freedom) with sample size $n = 100$ and $M = 500$ Monte Carlo replications.

4 Results

Monte Carlo study. The results for all 24 scenarios are summarised in Figs. 2 and 3; the full numeric table is provided in the supplementary material (see Data and Code Availability).

PMM3 significantly outperforms OLS at $\lambda = 1.5$: ARE ranging from 1.5 to 23.8 – the theory has been confirmed. The empirical efficiency coefficient $g_{3\text{emp}}$ approximates its theoretical value $g_{3\text{theor}}$ with an accuracy of approximately $\pm 10\%$ for sufficient sample sizes ($n = 100$). For the most challenging scenario, $\lambda = 5$, $n = 100$, the efficiencies were $\text{ARE}(\text{PMM3}) = 23.8$ compared with $\text{ARE}(\text{MLE-TN}) = 27.4$.

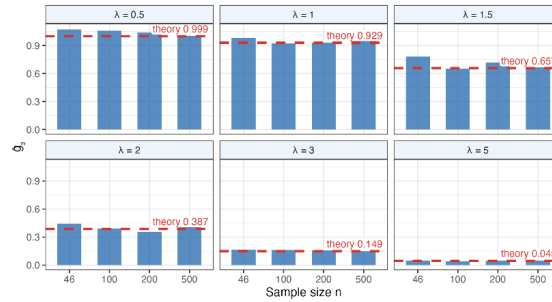


Fig. 2. Empirical $\hat{g}_3$ (bars) against theoretical g_3 (dashed line) for the 24 scenarios ($\lambda \times n$, $M = 500$).

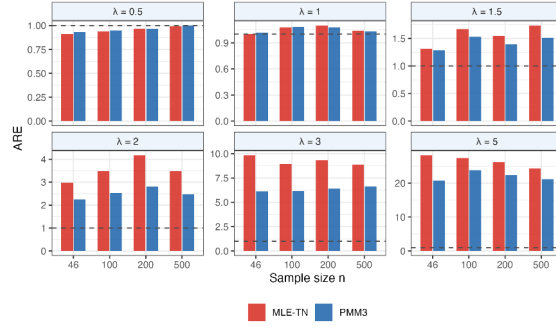


Fig. 3. Monte Carlo ARE of MLE-TN and PMM3 relative to OLS ($M = 500$); ARE > 1 means more efficient than OLS.

Real-world data. PMM3 achieved the smallest prediction error: LOO-MSE = 0.2022, which is lower than that obtained by OLS = 0.2070 and MLE-TN = 0.2109.

Bootstrap analysis indicates a 43% reduction in variance ($g_{3emp} = 0.570$), compared with the theoretical prediction of 51.7%. Given the relatively small sample size ($n = 50$), this agreement can be considered satisfactory.

Table 1. Comparison of estimation methods for iris versicolor ($n = 50$).

Method	$\hat{\beta}_1$	LOO-MSE	g_{3emp} (bootstrap)
OLS	0.865	0.2070	1.000 (baseline)
PMM3	0.797	0.2022	0.570
MLE-TN	0.753	0.2109	0.746

The relative variance is computed as $g_{3emp} = \text{Var}(\text{method})/\text{Var}(\text{OLS})$ using bootstrap estimation with $B = 2000$ replications.

Robustness under misspecification. PMM3 is robust: ARE ≥ 1.02 for all four distributions, so it performs no worse than OLS even when its assumptions fail; for the uniform law ($\gamma_4 = -1.2$) it reaches 2.685, above MLE-TN (1.655). MLE-TN is considerably more sensitive to misspecification: under the triangular law its ARE = 0.969 is below OLS. The Newton-Raphson implementation converged in 100% of the 200 replications for every $\lambda \in \{0.8, 1.0, 1.2, 1.5, 2.0, 3.0\}$, in 2.8-3.7 iterations on average, with absolute bias below 0.003.

Table 1. ARE under model misspecification ($n = 100, M = 500$).

True DGP	γ_4 (true)	ARE(MLE-TN)	ARE(PMM3)
Uniform	-1.200	1.655	2.685
Triangular	-0.600	0.969	1.061
Logistic	+1.200	0.994	1.023
Student's t10	+1.000	0.982	1.047

5 Discussion

When the TN model is correct and n is large, MLE-TN is the most efficient estimator, as the Cramér–Rao bound predicts; in the hardest scenario ($\lambda = 5, n = 100$) PMM3 lags behind it by only about 15% without any knowledge of the error density. Under misspecification the ordering changes (Table 2), so the choice between the two depends less on attainable efficiency than on how reliable the assumed error model is.

On the Iris data, despite its additional parameters $(\eta | \lambda)$, MLE-TN performed worst, which suggests that for datasets characterized by $n = 50$ and $\gamma_4 \approx -0.97$, PMM3 is the preferred estimator.

Negative example. For the whiteside dataset (MASS package, $n = 56$) the residuals give $\gamma_4 = -0.431$, i.e. above the threshold: the theoretical gain is only 8.0% ($g_{3\text{theor}} = 0.92$), the bootstrap ratio is $g_{3\text{emp}} = 0.978$ and LOO-MSE PMM3/OLS = 0.999. Five further datasets with $\gamma_4 \in [-0.54, -0.12]$ give $g_{3\text{emp}} \in [1.03, 1.36]$ and LOO-MSE PMM3/OLS ≤ 1.013 , i.e. no gain, which supports the threshold $\gamma_4 \notin -0.7$.

Recent two-piece regression models [3], [14], [18], [26] gain efficiency through a fully specified likelihood, whereas semiparametric proposals [10], [15], [19] relax the error density to limit the effect of misspecification. PMM3 belongs to the second group but needs only finite moments up to the sixth order. Limitations: the errors are assumed i.i.d. and symmetric, the simulations cover $n \leq 500$, and the threshold rests on seven datasets, so it is a rule of thumb rather than a fixed boundary.

6 Conclusions

1. Theoretical results. PMM3 provides a 7% to 95% reduction in variance relative to OLS for TN-distributed errors with $\lambda \in \{1, \dots, 5\}$. Under correct specification MLE-TN gains even more but estimates $p + 2$ parameters. The empirical efficiency coefficient g^3 of PMM3 converges to its theoretical value for sample sizes of approximately $n = 100 - 200$.

2. Robustness. Under misspecification PMM3 never falls below OLS (ARE 1.02–2.68), whereas MLE-TN can; PMM3 is the safe choice when the error distribution is uncertain.
3. Real-world data. For the Iris versicolor dataset ($\gamma_4 = -0.966$, $n = 50$) PMM3 reduces variance by 43% and gives the lowest LOO-MSE.
4. Empirical threshold. PMM3 gives a practical advantage over OLS only when $\gamma_4 \notin [-0.7, 0]$; for $-0.7 < \gamma_4 < 0$ the prediction error stays within 2% of OLS. Prefer MLE-TN when the TN model is justified and n is large, PMM3 when the error distribution is uncertain, and OLS for near-Gaussian errors.

Data and Code Availability. The R code of the study, results of all experiments and a script that reruns the simulation and checks it against the stored tables are openly available under the MIT licence at <https://github.com/SZabolotnii/Ku-PMM3-MLE-TN-code-supplement> and archived on Zenodo: <https://doi.org/10.5281/zenodo.22849684>.

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